

desktop. With these, the desktop can connect to the server via SSH using the private key. While connecting, the server sends a challenge to the desktop. As an answer, the desktop sends the challenge signed with the private key back to the server. The server can verify this signature with the public key from the desktop, which is located in the user's home directory on the server.

So if you want to access System B from System A without entering a password, follow the steps shown below.

Step 1: Create SSH public and private keys

From System A, you have to type the following command to generate key pairs under the user account:

```
user@A#ssh-keygen -t rsa
```

This will create the following two files under the user's home folder:

1. `/home/user/.ssh/id_rsa.pub`
2. `/home/user/.ssh/id_rsa`

The first file is the public key for the user and the second one is the private key.

Step 2: Set up the SSH public key on System B

On System B, you now have to copy the content of `id_rsa.pub` file, which you have created for the user on System A, to the following file under the home folder of user2 at System B:

```
/home/user2/.ssh/authorized_keys
```

Now all is set. You will have to restart your SSH server on System B and try to log in from System A to System B as the user:

```
user@A#ssh user@B
```

...and you will be logged in on system B without entering a password.



This is a very old and common tip but is required frequently, and hence has been suggested for new users to try out.

—Sanjay Adiwai, adlsanjay@gmail.com



Access Linux partitions and file systems in Windows (in a dual boot system)

Nowadays, many of us have started using open source OSs and applications. Hence, we dual boot, having installed both Windows and Linux on the same system. By default, we can access Windows partitions (NTFS and FAT based file systems) and can read/write data in Windows through Linux OSs. Sometimes, though, we might need to access Linux partitions from Windows in dual boot mode or for external storage devices (formatted in `ext3/ext4` file systems).

The free open source utility `ext2fsd` (<https://sourceforge.net/projects/ext2fsd/files/>) can do the job for you.

We need to install `ext2fsd` in Windows and run it for it to show all the partitions, including Linux partitions. To access the partition, right-click on it and choose 'Change Drive Letter'. This will automatically assign partition names like in Windows, and one can access the partition easily.

—Amit Doegar, amit@nitttrchd.ac.in



Manage multiple programs from a single terminal using `tmux`

When you want to run multiple programs, you need to open many windows, which can be tedious. So an alternate option is to use '`tmux`' (which stands for terminal multiplexer). As the name suggests, `tmux` gives us the power to use multiple terminals from a single screen. It can be detached and reattached from/to the screen when required.

Here is how to install `tmux`.

For Ubuntu/Debian systems:

```
$sudo apt update
$sudo apt install tmux
```

Then, run the following command to check for the successful installation of `tmux`:

```
$tmux -V
```

If successfully installed, you will get an output showing the `tmux` version, as follows:

```
tmux 2.1
```

You can install `tmux` on MacOS also, the instructions for which can be found online.

After installation, you can take help on how to use `tmux`, with the `man` command, as shown below:

```
$man tmux
```

—Akhilesh Pandey, akhil.pan.darj@gmail.com

END 

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